

Project 03: Prompt Chaining Studio for Summarization

Multi-Stage Sequential Prompt Transformation Pipeline

1. Executive Overview

The Prompt Chaining Studio implements a 6-stage sequential LLM processing pipeline. Each stage performs a specific transformation step (Text Cleaning -> Fact Extraction -> Outline Generation -> Draft Summarization -> Self-Critique -> Polished Output), tracking latency and token telemetry per stage.

2. System Architecture & Workflow Pipeline

- Stage 1: Clean & Normalize Input Raw Text
- Stage 2: Extract 3-5 Core Factual Bullet Points
- Stage 3: Generate Structured 3-Part Outline
- Stage 4: Draft Concise Summary
- Stage 5: Self-Critique & Completeness Scoring (0-100)
- Stage 6: Final Polished Executive Summary Output

3. API Endpoints Specification

Method	Endpoint	Description
GET	/	Serves Prompt Chaining Studio web UI
POST	/api/chain	Executes 6-stage prompt chain and returns stage outputs & telemetry

4. Deployment & Automated Verification

Standalone Uvicorn Launch:

cd 03_prompt_chaining/src && python -m uvicorn server:app --host 127.0.0.1 --port 8003

Automated Test Suite Execution:

python -m pytest 03_prompt_chaining/tests/